Beyond Weight Loss: GLP-1 Based Anti-Obesity Therapies and the Transformation of Chronic Disease Management

Running Title: GLP-1 Therapies and the Future of Obesity Care Revolution

Article Type: Editorial

Word Count: Approximately 1,800 words

Keywords: Obesity, GLP-1 receptor agonists, semaglutide, tirzepatide, cardiovascular outcomes, chronic disease management, precision medicine

---

Abstract

The emergence of glucagon-like peptide-1 (GLP-1) receptor agonists and dual incretin therapies has profoundly impacted obesity and related metabolic disease care. Obesity, long believed to reflect inadequate lifestyle selections, is now increasingly recognized as a chronic illness deserving of concentrated and multidisciplinary medical management. Landmark studies elucidating the cardiovascular outcome benefit of semaglutide and tirzepatide have unveiled unprecedented weight-loss capabilities and cardiometabolic assistances, transcending prior understanding of obesity therapy's scope and limits. The SELECT trial demonstrated that semaglutide's cardiovascular protective upshot in obesity with recognized cardiovascular disease without diabetes, whereas the recent SURMOUNT clinical development program highlighted the long-term weight conservation challenges with tirzepatide.

Amidst controversies regarding long-term safety, treatment affordability, healthcare equity, and the potential over-medicalization of obesity, this editorial aims to critically evaluate the current evidence surrounding incretin-based obesity therapies, explore their clinical implications, and anticipate future innovations in obesity care to chart the field's promising trajectory. Incretin-based therapies represent one of the most paradigm-shifting innovations in contemporary metabolic medicine, with far-reaching implications for chronic disease management and public health.

Introduction

Obesity has evolved into one of the most pressing public health crises globally. This condition is associated with increased risk of cardiovascular diseaserisk, type 2 diabetes mellitus, chronic kidney disease, obstructive sleep apnea, osteoarthritis,

several malignancies, and reduced life expectancy. Traditionally viewed as a consequence of, inadequate lifestyle choices, obesity is now understood as a complex chronic disease influenced by genetic, neurohormonal, environmental, behavioral, and socioeconomic determinants.

The development of GLP-1 receptor agonists and dual incretin therapies represents a major breakthrough in obesity management. Semaglutide and tirzepatide have demonstrated unprecedented value in attaining sustained weight loss that rivals bariatric surgery outcomes. These therapies have generated considerable excitement within the medical communal, pharmaceutical industry, and patient populations while simultaneously raising critical scientific and ethical questions.

The central thesis of this editorial is that incretin-based obesity therapies have fundamentally converted the treatment landscape, shifting the focus from symptomatic weight management to comprehensive cardiometabolic disease modification. However, their widespread adoption wants careful consideration of long-term implication, healthcare economics, and societal impact.

Analysis

A Paradigm Shift in Obesity Management

The clinical significance of modern incretin-based obesity therapies covers beyond more reduction. Semaglutide, in particular, demonstrated remarkable cardiovascular benefits.

The SELECT trial revealed that weekly semaglutide administration reduced major adverse cardiovascular actions among individuals with obesity and established cardiovascular disease without diabetes. The primary composite endpoint included cardiovascular death, nonfatal myocardial infarction, and nonfatal stroke. This finding has important implication, suggesting that obesity pharmacotherapy can directly impact clinically relevant outcomes rather than just surrogate markers.

Similarly, tirzepatide has generated considerable attention due to its dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 receptor agonism. The SURMOUNT program's finding inspiring body weight reduction in participants with obesity or overweight. Long-term extension have more emphasized the importance of sustained treatment in maintaining weight loss benefits.

Beyond Weight Loss

The biological mechanisms of incretin therapies encompass a range of physiological processes, including:

- Central appetite suppression

- Enhanced satiety signals
- Delayed gastric emptying
- Enhanced insulin sensitivity
- Reduction of inflammatory activity
- Cardiometabolic risk factor improvement

These mechanisms may explain the expanding interest in their application across conditions such as heart failure, chronic kidney disease, metabolic dysfunction-associated steatotic liver disease (MASLD), and obstructive sleep apnea.

The therapeutic objective is therefore evolving. Rather than directing exclusively on body weight, clinicians increasingly consider broader outcomes including cardiovascular protection, metabolic enhancement, quality of life, and reduction of obesity-related complications.

Public Health Implications

Obesity imposes a substantial burden on healthcare systems globally. Effective pharmacological interventions can significantly reduce long-term healthcare expenditures by preventing obesity-related complications. This economic perspective has prompted governments, insurers, and healthcare organizations to reassess reimbursement strategies and treatment algorithms. The potential to prevent cardiovascular events and other long-lasting disease complications provides a compelling economic rationale for expanding access to obesity therapies.

Critical Evaluation

Long-Term Safety Uncertainty

Despite encouraging results, long-term evidence remains limited. Most major clinical trials have follow-up periods measured in years rather than decades.

Important unanswered questions include:

- Long-term cardiovascular safety
- Effects of prolonged exposure
- Sustained adherence
- Rare adverse events
- Outcomes in special populations

Continued post-marketing surveillance and long-term observational studies will therefore be essential.

Weight Regain and Chronic Therapy

One of the most important observations emerging from obesity pharmacotherapy research is the tendency toward weight regain after treatment discontinuation.

This finding suggests that obesity treatment may resemble management of hypertension, diabetes, or hyperlipidemia, where ongoing therapy is often required. Such a model raises queries about sustainability, healthcare costs, and patient expectations.

The concept of lifelong obesity medication remnants controversial and requires careful evaluation from both clinical and economic perspectives.

Affordability and Access

The high charge of incretin therapies creates a substantial barrier.

Patients in low- and middle-income countries may experience limited access. Even within high-income healthcare systems, insurance treatment may be inconsistent.

If these treatments continue to demonstrate significant reductions in cardiovascular events and chronic disease burden, policymakers will need to determine how to balance innovation with affordability.

Risk of Over-Medicalization

The impressive effectiveness of these therapies has contributed to extensive media attention. In some settings, public discourse has shifted toward cosmetic weight loss rather than evidence-based obesity treatment.

Excessive focus on pharmacological interventions risks minimizing the importance of:

- Behavioral modification
- Nutritional intervention
- Physical activity
- Population-level prevention strategies

Medication would complement rather than replace comprehensive obesity management.

Ethical and Social Considerations

Obesity is strongly influenced by environmental and socioeconomic determinants.

Access to nutritious food, safe living situations, education, healthcare services, and opportunities for physical activity continue to shape obesity prevalence.

The availability of current medications should not divert attention from addressing these structural determinants. Failure to do so may widen health disparities despite scientific progress.

Future Directions

Next-Generation Therapeutics

The next decade is likely to produce increasingly sophisticated anti-obesity therapies.

Research efforts are currently exploring combinations involving:

- GLP-1
- GIP
- Glucagon
- Amylin
- Additional metabolic targets

Future agents may achieve greater efficacy while improving tolerability.

Precision Obesity Medicine

Precision medicine represents an especially promising direction.

Potential innovations include:

- Genomic profiling
- Biomarker-guided therapy
- Metabolic phenotyping
- Individualized treatment algorithms

These policies may improve treatment selection and optimize outcomes.

Longer-Term Clinical Evidence

Future study should prioritize:

1. Extended follow-up studies.
2. Comparative effectiveness trials.
3. Diverse population enrollment.

4. Real-world evidence studies.
5. Health equity assessments.

Such investigations will strengthen the indication base supporting widespread implementation.

Health Economics Research

Policy conclusions increasingly require evidence regarding:

- Cost-effectiveness
- Budget impact
- Reimbursement strategies
- Population health outcomes

If obesity therapies consistently demonstrate discounts in cardiovascular actions and hospitalizations, economic analyses may support broader healthcare coverage.

Integrated Care Models

The future of obesity management will likely involve multidisciplinary care that combines:

- Pharmacotherapy
- Nutrition counselling
- Physical activity programs
- Behavioral interventions
- Psychological support

The greatest clinical benefit is expected from integrated approaches rather than isolated treatment modalities.

Conclusion

GLP-1 receptor agonists and dual incretin therapies have really revolutionized the treatment of obesity. Semaglutide and tirzepatide have demonstrated extraordinary efficacy in achieving significant and sustained weight loss. The SELECT trial demonstrated the cardiovascular benefit of semaglutide in patients with obesity and pre-existing cardiovascular disease, while the SURMOUNT program has provided valuable insights into the long-term weight maintenance with tirzepatide.

Nevertheless, critical issues involving safety, affordability, equitable access, and lifelong treatment requirements remain unresolved. As the field evolves, researchers, clinicians, policymakers, and healthcare organizations must work collaboratively to ensure that scientific innovation translates into meaningful and equitable health outcomes.

The future of obesity medicine will depend not only on increasingly effective therapies but also on the ability of healthcare systems to integrate those therapies within comprehensive, patient-centered models of chronic disease management.

References :

1. Lincoff AM, Brown-Frandsen K, Colhoun HM, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. *N Engl J Med*. 2023;389:2221-2232. doi:10.1056/NEJMoa2307563.
2. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387:205-216. doi:10.1056/NEJMoa2206038.
3. Wilding JPH, Batterham RL, Calanna S, et al. Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med*. 2021;384:989-1002. doi:10.1056/NEJMoa2032183.
4. Rubino D, Abrahamsson N, Davies M, et al. Effect of continued weekly subcutaneous semaglutide vs placebo on weight loss maintenance in adults with overweight or obesity: The STEP 4 randomized clinical trial. *JAMA*. 2021;325(14):1414-1425. doi:10.1001/jama.2021.3224.
5. Horn DB, Aronne LJ, Wharton S, et al. Tirzepatide for maintenance of bodyweight reduction in people with obesity in the USA (SURMOUNT-MAINTAIN): a multicentre, double-blind, randomised, placebo-controlled trial. *Lancet*. 2026. DOI pending publication.
6. Drucker DJ. Mechanisms of action and therapeutic application of glucagon-like peptide-1. *Cell Metab*. 2018;27(4):740-756. doi:10.1016/j.cmet.2018.03.001.
7. Blüher M. Obesity: global epidemiology and pathogenesis. *Nat Rev Endocrinol*. 2019;15(5):288-298. doi:10.1038/s41574-019-0176-8.
8. Bray GA, Kim KK, Wilding JPH. Obesity: a chronic relapsing progressive disease process. *Obes Rev*. 2017;18(7):715-723. doi:10.1111/obr.12551.
9. Garvey WT, Mechanick JI, Brett EM, et al. American Association of Clinical Endocrinologists and American College of Endocrinology comprehensive clinical

practice guidelines for medical care of patients with obesity. *Endocr Pract*. 2016;22(suppl 3):1-203. doi:10.4158/EP161365.GL.

10. Yumuk V, Tsigos C, Fried M, et al. European guidelines for obesity management in adults. *Obes Facts*. 2015;8(6):402-424. doi:10.1159/000442721.

11. Solomon SD, McMurray JJV, Claggett B, et al. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2023;389:1069-1084. doi:10.1056/NEJMoa2306963.

12. Newsome PN, Buchholtz K, Cusi K, et al. A placebo-controlled trial of subcutaneous semaglutide in nonalcoholic steatohepatitis. *N Engl J Med*. 2021;384:1113-1124. doi:10.1056/NEJMoa2028395.

13. Mann JFE, Ørsted DD, Brown-Frandsen K, et al. Liraglutide and renal outcomes in type 2 diabetes. *N Engl J Med*. 2017;377:839-848. doi:10.1056/NEJMoa1616011.

14. Blackman A, Foster GD, Zammit G, et al. Effect of liraglutide 3.0 mg in individuals with obesity and moderate or severe obstructive sleep apnea. *Int J Obes (Lond)*. 2016;40(8):1310-1319. doi:10.1038/ijo.2016.52.

15. Acosta A, Streett S, Kroh MD, et al. White paper AGA: power of personalized medicine in obesity treatment. *Clin Gastroenterol Hepatol*. 2021;19(11):2231-2241. doi:10.1016/j.cgh.2021.04.041.

16. Cawley J, Biener A, Meyerhoefer C, et al. Direct medical costs of obesity in the United States and the most populous states. *J Health Econ*. 2021;77:102378. doi:10.1016/j.jhealeco.2021.102378.

17. Wharton S, Astrup A, Endahl L, et al. Estimating treatment effects of semaglutide and lifestyle intervention on obesity-related complications. *Obesity (Silver Spring)*. 2022;30(7):1323-1332. doi:10.1002/oby.23462.

18. Koliaki C, Spinos T, Spinou M, et al. Defining the optimal dietary approach for safe, effective and sustainable weight loss in overweight and obese adults. *Healthcare (Basel)*. 2018;6(3):73. doi:10.3390/healthcare6030073.